Air Pollution and the Air Quality Index

Knowledge document 038 of 100

Introduction to Air Pollution

Air pollution is the presence in the air of substances in amounts that harm human health, living things and the environment. These substances can be solid particles, liquid droplets or gases. Air pollution comes from both natural sources, such as dust storms, wildfires and volcanoes, and human activities. The World Health Organization estimates that around 7 million deaths each year worldwide are linked to air pollution, indoor and outdoor combined, and it is a major risk factor for asthma, heart disease, stroke, lung cancer and lung infections. India has some of the most polluted cities in the world, so understanding air quality is important for every citizen.

Major Air Pollutants

The main outdoor pollutants are particulate matter (PM), which is a mixture of tiny solid and liquid particles; nitrogen dioxide (NO2), mostly from vehicles; sulphur dioxide (SO2), from burning coal and oil; carbon monoxide (CO), from incomplete burning of fuels; ground-level ozone (O3), formed by sunlight acting on other pollutants; ammonia, lead and volatile organic compounds. Indoor air pollution comes from cooking with firewood, cow dung or coal, which affects many rural households, and from tobacco smoke.

What Is PM2.5?

PM2.5 is particulate matter with a diameter of 2.5 micrometres or less. A micrometre is one-thousandth of a millimetre. PM2.5 particles are about 30 times thinner than a human hair, which is about 70 micrometres wide, so they are invisible individually. PM10 refers to coarser particles up to 10 micrometres in diameter, such as dust. Because PM2.5 particles are so small, they can be breathed deep into the lungs, pass into the bloodstream, and damage the lungs, heart and brain. PM2.5 is considered one of the most harmful forms of air pollution. It is produced by vehicle exhaust, burning of coal, biomass and waste, construction dust, industrial emissions, crop residue burning, and by chemical reactions of gases in the air. The World Health Organization's 2021 guideline recommends that annual average PM2.5 should not exceed 5 micrograms per cubic metre, while India's national standard is 40 micrograms per cubic metre annually and 60 micrograms per cubic metre over 24 hours.

What Is the Air Quality Index?

The Air Quality Index (AQI) is a number scale that converts complex measurements of several pollutants into a single number and colour category that explains how clean or polluted the air is and what health effects may be expected. India's National Air Quality Index was launched on 17 October 2014 by Prime Minister Narendra Modi, and it considers eight pollutants: PM10, PM2.5, nitrogen dioxide, sulphur dioxide, carbon monoxide, ozone, ammonia and lead. The index runs from 0 to 500 and has six categories. An AQI of 0 to 50 is Good, 51 to 100 is Satisfactory, 101 to 200 is Moderate, 201 to 300 is Poor, 301 to 400 is Very Poor, and 401 to 500 is Severe.

What Does an AQI Above 300 Mean?

An AQI above 300 is in the "Very Poor" category (301 to 400), and above 400 is "Severe". An AQI in the Very Poor range means that the air is highly polluted and may cause respiratory illness in people on prolonged exposure. In the Severe category, above 400, the air affects even healthy people and seriously affects those with existing diseases such as asthma and heart problems. During such days, people are advised to avoid outdoor activity, especially children, the elderly and patients, to keep windows closed, wear N95-type masks outdoors, and use air purifiers indoors. Schools may be closed and construction may be stopped under emergency measures such as the Graded Response Action Plan (GRAP) in the Delhi region, where the AQI frequently crosses 400 in winter.

Two Main Sources of Air Pollution in Cities

Two main sources of air pollution in cities are vehicle exhaust and construction and road dust. Cars, trucks, buses and two-wheelers burn petrol and diesel and release particulates, nitrogen oxides and carbon monoxide, while congested traffic makes it worse. Other major sources include industrial emissions, coal-fired power plants, burning of garbage, diesel generators, and in the north Indian winter, the burning of crop stubble in Punjab and Haryana combined with cold, still air that traps pollutants. Weather also matters: temperature inversion in winter prevents pollutants from rising and dispersing.

Air Quality in Chennai and Tamil Nadu

Chennai generally has better air quality than Delhi because sea breezes help disperse pollutants, though levels rise near industrial areas like Manali and Ennore, and on busy roads, during Deepavali firecracker use and during Bhogi bonfires in January, when burning of old materials increases pollution. The Tamil Nadu Pollution Control Board runs air-quality monitoring stations across the state.

Policies and Solutions

The Air (Prevention and Control of Pollution) Act was passed in 1981, and the Central Pollution Control Board, set up in 1974, monitors air quality. The National Clean Air Programme (NCAP), launched in January 2019, targets reducing particulate pollution in 130 cities. India moved to Bharat Stage VI (BS-VI) emission norms for vehicles in April 2020, skipping BS-V. Other measures include electric vehicles, cleaner cooking fuel under the Pradhan Mantri Ujjwala Yojana, metro rail, renewable energy, dust control at construction sites, tree planting, and curbing stubble burning. Individuals can help by using public transport, carpooling, avoiding burning waste and checking the AQI on apps like SAMEER before outdoor activity.

Challenges and Future

Challenges include the rapid growth of vehicles, dependence on coal, weak enforcement and cross-boundary pollution. Low-cost sensors, satellite monitoring and AI forecasts are improving the ability to manage air quality.

Quick Facts

- PM2.5 is particulate matter of 2.5 micrometres or smaller that can enter the lungs and bloodstream.
- India's AQI was launched on 17 October 2014 and has six categories from Good to Severe.
- An AQI above 300 means Very Poor air that may cause respiratory illness; above 400 is Severe.
- Vehicle exhaust and construction dust are two major urban pollution sources.
- India moved to BS-VI vehicle emission norms in April 2020.